

345. Reverse Vowels of a String

Solved 

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Given a string `s`, reverse only all the vowels in the string and return it.

The vowels are `'a'`, `'e'`, `'i'`, `'o'`, and `'u'`, and they can appear in both lower and upper cases, more than once.

Example 1:

Input: `s = "IceCreAm"`

Output: `"AceCreIm"`

Explanation:

The vowels in `s` are `['I', 'e', 'e', 'A']`. On reversing the vowels, `s` becomes `"AceCreIm"`.

Example 2:

Input: `s = "leetcode"`

Output: `"leotcede"`

Constraints:

- `1 <= s.length <= 3 * 105`
- `s` consist of **printable ASCII** characters.

```
class Solution {
    public String reverseVowels(String s) {
        StringBuilder sb = new StringBuilder(s);
        int left = 0;
        int right = sb.length() - 1;
        while (left < right) {
            while (left < right && !isVowel(sb.charAt(left))) {
                left++;
            }
            while (left < right && !isVowel(sb.charAt(right))) {
                right--;
            }
            if (left < right) {
                char temp = sb.charAt(left);
                sb.setCharAt(left, sb.charAt(right));
                sb.setCharAt(right, temp);
                left++;
                right--;
            }
        }
        return sb.toString();
    }
    private boolean isVowel(char ch) {
        return ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u'
            || ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U';
    }
}
```

→ Set $left = 0$, $right = n - 1$

$O(N)$

→ while $left < right$

* Increment $left$ until it is less than $right$ & is a vowel.

* Same for $right$ (decrement)

* If $left < right$, then swap & $left++$,
 $right--$.